

BRAC University, Dhaka
Department of Computer Science and Engineering

CSE 446: Blockchain and Cryptocurrencies, Spring 2026, Quiz - 3

Marks: 20

Time: 20 Min

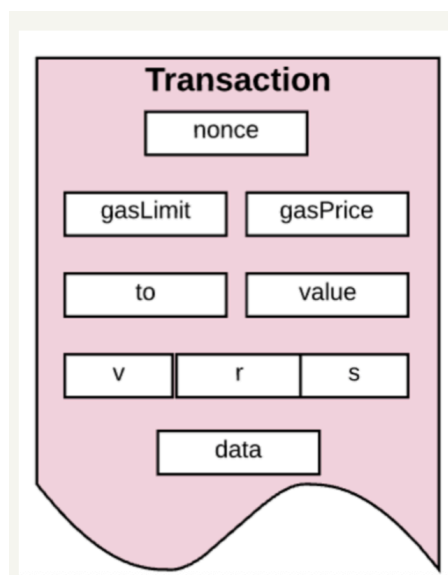
Name:	ID:	Section:
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1. Determine if the following sentences are true or false. For any false sentence, write its correct form. 4*1.5 = 6

- i. An external transaction is stored in the blockchain. (T)
- ii. PoS Ethereum has forks. (F)
PoS Ethereum has forks.
- iii. An internal transaction is signed by the smart-contract (F)
A smart-contract cannot sign any tx as it has no private key.
- iv. Fees in a transaction are collected by the transaction receiver. (F)
Fees are collected by the miners/stakers.

2. Fill in the blanks 4*1.5 = 6

- i. The address of a contract account is generated by __concatenating sender address and nonce and then serializing using RLP and then hashing__
- ii. Two components of an Ethereum tx are _____ & _____
Any two from the following tx structure figure



- iii. Fees in Ethereum are calculated by __gasLimit x gasPrice__
- iv. Examples of two external transactions are __contract creation transaction__ and __payment/contract executing tx__

3. Briefly answer the following questions.

4*2 = 8

- i. Write two differentiating characteristics of Ethereum blockchain with respect to Bitcoin blockchain.

Ethereum blockchain has four trees while Bitcoin blockchain has one tree.

Ethereum blockchain stores the state of the smart-contract while Bitcoin blockchain has no smart-contract.

- ii. How can we know if a certain Ethereum address is a contract account address or an EOA address?

By examining the state trie of respective address, if the state tries stores only two pieces of information for an address, that address belongs to an EOA, if there are four pieces of information, then the address belongs to a contract account.

- iii. Why is the absence of penalty is an issue for PoW? How does PoS solve this issue?

The absence of penalty enables a miner to misbehave. A miner in PoS needs to stake certain cryptocurrencies in the escrow account. If the minder is found to misbehave, a penalty is imposed.

- iv. Draw an MPT for the account 11AE05 containing 1.25 ETH. Consider that the first two characters are the shared nibble in the root extension node.

Extension Node		
prefix	shared nibble	next node
0	11	

Branch Node															
0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Leaf Node		
prefix	key-end	value
3	E05	0.125 ETH